SUPPORTING ADULT GUIDANCE

Expanding and factorising

Algebraic Expressions and Identities · Lesson 2. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson treats expanding and factorising as one rectangle read two ways. In the main activity, Expand tasks give the sides and learners fill the cells; Factorise tasks give only the total, and learners choose the sides while the cells update live until the totals match. The tasks include common factors, negative numbers, a perfect square, the difference of two squares, and a two-step stretch. Learners then use a factor-pair finder, and find every value that makes a quadratic factorise.

Driving Question: *How can one expression be written as a sum and as a product?*

The mathematics, in plain terms

$(x + p)(x + q) \equiv x^2 + (p + q)x + pq$, so $x^2 + bx + c$ factorises over the integers exactly when some factor pair of c sums to b. The difference of two squares is the case $b = 0$. 'Fully factorised' means the highest common factor has been taken out and no bracket factorises further. Factorising $ax^2 + bx + c$ with $a \neq 1$ (other than by a common factor) is left for the Summer Quadratics unit.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Expanding with the grid

Expand and simplify $(2x + 3)(x - 4)$.

$(2x + 3)(x - 4)$

×	x	-4
$2x$	$2x^2$	$-8x$
3	$3x$	-12

1. Put one bracket down the side and the other as the column headings.	2 by 2 grid
2. Fill each cell: its row term × its column term, keeping the signs.	$2x^2$, $-8x$, $3x$, -12
3. Collect the x terms (the shaded cells).	$-8x + 3x = -5x$
4. Write the sum.	$2x^2 - 5x - 12$

$(2x + 3)(x - 4) = 2x^2 - 5x - 12$.

Example 2 — Factorising: find the sides

Factorise $x^2 - x - 12$.

The rectangle for $(x + 3)(x - 4)$

×	x	-4
x	x^2	$-4x$
3	$3x$	-12

1. The last cell must be -12, and the two shaded cells must add to $-x$.	multiply to -12, add to -1
2. Try the factor pairs of -12: 3 and -4.	$3 \times (-4) = -12$, $3 + (-4) = -1$
3. These are the sides of the rectangle.	$(x + 3)(x - 4)$
4. Check by expanding: the grid gives $x^2 - 4x + 3x - 12$.	$= x^2 - x - 12$ ✓

$x^2 - x - 12 = (x + 3)(x - 4)$.

One rectangle, both ways

Year 9 met the area model for simple brackets. Here the same grid runs backwards: learners choose the sides and watch the cells change, so 'multiply to *c*, add to *b*' is seen before it is stated, and the stretch proves it. The factor-pair finder makes the search systematic, and the Creativity puzzle shows why the factor pairs tell you exactly how many answers there are.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Sum or product? Recognise expanded, factorised and partly factorised forms.	~5 min
Movement	The two-way area grid: expand from the sides; factorise from the total. Stretch: prove the identity behind the method.	~17 min
Reflection	The factor-pair finder: multiply to *c*, add to *b*.	~9 min
Creativity	Which ones factorise? Find every value, and know when you have them all.	~9 min
Rest	12 as 3 × 4 or 12 × 1: the middle term chooses.	~2 min

Nutrition	Intellectual: flexibility — seeing one thing two ways.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Missing the middle terms: $(x + 5)(x - 2) = x^2 - 10$.
Ask: “How many cells does the grid have?”

Sign errors with negatives in the grid.
Ask: “What is a positive times a negative?” Write the signs in the headers.

Taking out a common factor that is not the highest: $2x(4x + 6)$.
Ask: “Does the bracket still have a common factor?”

Choosing numbers that add to *c* and multiply to *b*.
Ask: “Which cell of the grid is the number term?”

Thinking $x^2 - 25$ cannot be factorised because there is no x term.
Ask: “What two numbers add to 0?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: expanding a bracket, factor pairs, and multiplying negative numbers. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner factorises $x^2 - 3x - 10$ as $(x + 5)(x - 2)$. The aim is a question that lets them find and fix the slip.

- “Expand your answer. What do you get?”
- “Which part is right, and which is not?”

- “What happens if you swap the signs?”
- “How could you check before expanding?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Should learners use the grid or a mnemonic like FOIL?
The grid. It works for any size of bracket (three brackets come next lesson) and shows why every term is needed.

Is $2(x^2 - x - 12)$ fully factorised?
Not yet: the bracket factorises further, to $(x - 4)(x + 3)$. The lesson recognises the value and asks for the full factorisation.

Is the proof required?
It is a stretch. The secure core is expanding with the grid and factorising with factor pairs.

What comes next?
Lesson 3: expanding three brackets.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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